

Supervised Linear Dimensionality Reduction Based on Factor Analysis and Information-Theoretic Criteria

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Abstract—Recently supervised linear dimensionality reduction method based on Information-Theoretic Criteria has attracted more attention. In existing methods, the first and second order statistics of the data in the original high-dimensional space is estimated, and then the transformation matrix based on the information-theoretic criteria is designed. However, it is difficult to accurately estimate these statistics in the original high-dimensional space, especially in the case of limited data. Thus the transformation matrix obtained may be nonoptimal, which will affect the final classification performance. To solve this problem, a novel supervised linear dimensionality reduction method is proposed in this paper. In our method, the statistical structure of the transformed low-dimensional subspace is described via factor analysis (FA) model, while the mutual information (MI) between the transformed data and their class labels is maximized. Moreover, the joint optimization of MI function and log-likelihood function is achieved, which can not only reduce the estimation errors, but also ensure the separability of the transformed data. Experimental results on benchmark datasets and radar data demonstrate the effectiveness of our method.

Keywords—supervised linear dimensionality reduction, mutual information, Factor Analysis, classification.

I. INTRODUCTION

With the increasing of large-scale high-dimensional data, dimensionality reduction is becoming more and more important for high-dimensional data classification. Typical linear dimensionality reduction methods include Linear Discriminant Analysis (LDA) [1-2], Principal Component Analysis (PCA) [3], Independent Component Analysis (ICA) [4-9], and Information Discriminant Analysis (IDA) [10].

The transformed matrix is learned in LDA via maximizing the between-class scattering while minimizing the within-class scattering of the transformed data [1]. However, the dimensionality of the transformed space in LDA can only be less than the number of data classes, which greatly restricts its performance and applications. To solve this problem, some linear dimensionality reduction methods based on the information-theoretic criteria are proposed [11-12]. The transformation matrix is designed in [11] by maximizing the mutual information (MI) between the transformed data and their class labels. Unfortunately, the MI is very difficult to calculate directly for non-Gaussian distributions. To overcome this disadvantage, IDA [10] and the work of mutual information maximization (MIM) [12] are proposed based on the information-theoretic criteria. IDA approximates the entropy of the Gaussian mixture model

(GMM) with the entropy of a global Gaussian distribution, and provides an approximate expression to the Shannon MI. In MIM, an analytic and explicit expression for the gradient of the Shannon MI with respect to the transformation matrix is given for any distribution, thus the Shannon MI can be maximized directly without any approximation. Both methods estimate the first and second order statistics of the data in the high-dimensional space, and then design the transformation matrix based on the information-theoretic criteria. However, it is difficult to accurately estimate these statistics in the original high-dimensional space, especially in the small sample size. Thus the transformation matrix obtained may be non-optimal, which will affect the classification performance.

To solve this disadvantage, a novel supervised linear dimensionality reduction method based on factor analysis (FA) model and MI is proposed in this paper. The proposed method is conveniently called maximizing mutual information of factor analysis (MMI-FA). In MMI-FA, the statistical structure of the transformed low-dimensional subspace is described via factor analysis model, while the mutual information (MI) between the transformed data and their class labels is maximized. Moreover, the joint optimization of MI function and log-likelihood function is achieved, which can not only reduce the estimation errors, but also ensure the separability of the transformed data.

The remainder of the paper is organized as follows: Section II introduces the model of MMI-FA. The performance of our method on Benchmark datasets and radar data is validated Section III. We summarize the paper by presenting some concluding remarks in Section IV.

II. MAXIMIZING MUTUAL INFORMATION OF FACTOR ANALYSIS

Traditional supervised dimensionality reduction methods based on MI directly estimate statistics in the original high-dimensional space, which may lead to inaccurate estimation of statistics, especially in the small sample size. To solve this problem, a new supervised dimensionality reduction method based on FA and information-theoretic criterion is presented in this paper. The basic idea will be described in the following.

Assuming that the high-dimensional data from class k is generated via the following FA model

$$\mathbf{x} = \mathbf{A}\mathbf{y} + \boldsymbol{\varepsilon} \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^{d \times 1}$ represents the original data from class k ,

This work was partially supported by the National Science Foundation of China (No.61771362, No. U1833203, No.61671354), and 111 Project (B18039).

$\mathbf{A} \in \mathbb{R}^{d \times m}$ is the transformation matrix with $m < d$, $\mathbf{y} \in \mathbb{R}^{m \times 1}$ denotes the low-dimensional latent feature of $\mathbf{x}$, and $\boldsymbol{\varepsilon} \in \mathbb{R}^{d \times 1}$ is the measurement noise. Thus the original high-dimensional data $\mathbf{x}$ can be viewed as the linear mapping from a latent feature $\mathbf{y}$ with the added noise $\boldsymbol{\varepsilon}$. Here we assume Gaussian prior on $\mathbf{y}$ and $\boldsymbol{\varepsilon}$, i.e., $p(\mathbf{y}|k) = \mathcal{N}(\mathbf{y}; \boldsymbol{\theta}, \mathbf{I})$ and $p(\boldsymbol{\varepsilon}|k) = \mathcal{N}(\boldsymbol{\varepsilon}; \boldsymbol{\theta}, \boldsymbol{\Psi}_k)$, where $\boldsymbol{\Psi}_k$ is a diagonal matrix, representing the noise variance on class k . We introduce such a FA model for each class of the data, in which the transformation matrix $\mathbf{A}$ is shared for all class, but the measurement noises' variance matrixes are different for the different classes.

For convenience, we define $\boldsymbol{\Psi} = \{\boldsymbol{\Psi}_k\}_{k=1}^K$. We want to find the model parameters $\{\mathbf{A}, \boldsymbol{\Psi}\}$ which can not only ensure the separability of the latent features but also describe the statistical structure of observed data accurately. That is the following optimization problem

$$\max_{\mathbf{A}, \boldsymbol{\Psi}} I_{\mathbf{A}, \boldsymbol{\Psi}}(C, Y) + \lambda L_{\mathbf{A}, \boldsymbol{\Psi}}(X) \quad (2)$$

where λ denotes the weighting factor, X denotes the observed dataset, $I_{\mathbf{A}, \boldsymbol{\Psi}}(C, Y)$ represents the MI function between latent feature variable Y and class variable C , $L_{\mathbf{A}, \boldsymbol{\Psi}}(X)$ represents the log-likelihood function of the linear statistical model. Expressions for log-likelihood function and MI function are described in the following.

According to the definition of statistical model and the properties of Gaussian distribution, the prior distribution of the original data from class k can be expressed as $p(\mathbf{x}|k) = \mathcal{N}(\mathbf{x}; \boldsymbol{\theta}, \mathbf{F}_k^{-1})$ with $\mathbf{F}_k = (\mathbf{A}\mathbf{A}^T + \boldsymbol{\Psi}_k)^{-1}$ and $k = 1, 2, \dots, K$. Thus the log-likelihood function of the entire dataset can be expressed as

$$\begin{aligned} L_{\mathbf{A}, \boldsymbol{\Psi}}(X) &= \sum_{k=1}^K \sum_{i=1}^{N_k} \ln P(\mathbf{x}_{ki}) \\ &= -\frac{1}{2} \sum_{k=1}^K N_k \left\{ d \ln(2\pi) + \ln |\mathbf{F}_k^{-1}| \right. \\ &\quad \left. + \text{tr} \left[\mathbf{F}_k \left(\frac{1}{N_k} \sum_{i=1}^{N_k} \mathbf{x}_{ki} \mathbf{x}_{ki}^T \right) \right] \right\} \end{aligned} \quad (3)$$

where $\mathbf{x}_{ki}$ represents the i th data from class k , and N_k is the data size for class k . According to the model definition, the latent feature $\mathbf{y}_{ki}$ satisfies the conditional distribution $p(\mathbf{y}_{ki} | \mathbf{x}_{ki}) = \mathcal{N}(\mathbf{y}_{ki}; \mathbf{A}^T \mathbf{F}_k \mathbf{x}_{ki}, \mathbf{I}_m - \mathbf{A}^T \mathbf{F}_k \mathbf{A})$. Given $\mathbf{x}_{ki}$, the latent feature $\mathbf{y}_{ki}$ can be approximated with a conditional mean $\mathbf{y}_{ki} = \mathbf{A}^T \mathbf{F}_k \mathbf{x}_{ki}$. Then the posterior distribution $q(\mathbf{y}|k) = \mathcal{N}(\mathbf{y}; \boldsymbol{\mu}_{y_k}, \boldsymbol{\Sigma}_{y_k})$ with

$$\boldsymbol{\mu}_{y_k} = \mathbf{A}^T \mathbf{F}_k \boldsymbol{\mu}_{x_k}, \boldsymbol{\Sigma}_{y_k} = \mathbf{A}^T \mathbf{F}_k \boldsymbol{\Sigma}_{x_k} \mathbf{F}_k \mathbf{A} \quad (4)$$

$$\text{Then } q(\mathbf{y}) = \sum_{k=1}^K w_k p(\mathbf{y}|k).$$

Similar to the IDA, we approximate the entropy of $q(\mathbf{y})$ with a global Gaussian entropy. Thus the MI under the linear statistical model can be expressed as

$$\begin{aligned} I_{\mathbf{A}, \boldsymbol{\Psi}}(C, Y) &= H(Y) - \sum_{k=1}^K w_k H(Y|k) \\ &= \frac{1}{2} \left[\log |\mathbf{A}^T \boldsymbol{\Omega} \mathbf{A}| - \sum_{k=1}^K w_k \log |\mathbf{A}^T \boldsymbol{\Omega}_k \mathbf{A}| \right] \end{aligned} \quad (5)$$

where $\boldsymbol{\Omega} = \sum_{k=1}^K w_k [\boldsymbol{\Omega}_k + (\boldsymbol{\mu}_k - \boldsymbol{\mu})(\boldsymbol{\mu}_k - \boldsymbol{\mu})^T]$, $\boldsymbol{\mu} = \sum_{k=1}^K w_k \boldsymbol{\mu}_k$, $\boldsymbol{\mu}_k = \mathbf{F}_k \boldsymbol{\mu}_{x_k}$, and $\boldsymbol{\Omega}_k = \mathbf{F}_k \boldsymbol{\Sigma}_{x_k} \mathbf{F}_k$. $\boldsymbol{\mu}_{x_k}$ and $\boldsymbol{\Sigma}_{x_k}$ are the prior mean and the prior covariance matrix for the original data from class k . The gradient functions with respect to parameters $\{\mathbf{A}, \boldsymbol{\Psi}\}$ can be derived based on (2), (3) and (5).

When we can obtain the low-dimensional feature $\mathbf{y}$ under the optimal model parameters $\{\mathbf{A}^*, \boldsymbol{\Psi}^*\}$, and further estimate the posterior distribution $q(\mathbf{y}|k)$ in the training stage. For a test data $\mathbf{x}^*$, we first calculate the low-dimensional feature $\{\mathbf{y}_k^*, \mathbf{y}_k^* = \mathbf{A}^{*T} (\mathbf{A}^* \mathbf{A}^{*T} + \boldsymbol{\Psi}_k^*)^{-1} \mathbf{x}^*\}_{k=1}^K$ under each linear statistical model, then classify the test data via the Bayesian classifier [13] as

$$t^* = \arg \max_{k=1, 2, \dots, K} q(\mathbf{y}_k^* | k) w_k \quad (6)$$

So far, we implement the entire supervised dimensionality reduction model, and the model parameters are also estimated. In practice, the classification task can be performed in two stages: training and test stages. The specific classification block diagram is shown in Fig. 1.

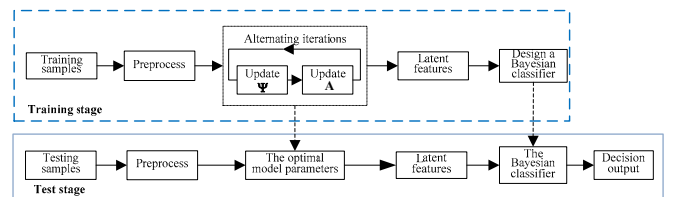


Fig. 1. The classification block diagram of the proposed method.

III. EXPERIMENTS AND RESULTS

A. Benchmark Datasets

The performance of the proposed method is evaluated experimentally with two benchmark datasets: Heart and Satellite. A detailed description is summarized in Table 1.

TABLE I. DESCRIPTION OF THE TWO DATASETS

Dataset	Number of Classes	Data Dimension	Data Size
Heart	2	13	270
Satellite	6	36	4435/2000

For Satellite dataset, training and test sets have been specifically designed by its donor, of which there are 4435 training data and 2000 test data in the Satellite dataset. For Heart dataset, we randomly split the dataset into two non-

overlapping groups of equal size, then select one group as training set and the other one as test set. Therefore, the model parameters can be estimated from the training set and validated on the test set. To control the variability caused by random partitions, the splitting work is repeated 20 times and the results are averaged over the 20 partitions.

Five related methods are considered to compare against the proposed method: FA, PCA, LDA, IDA, and MIM. The performance is evaluated using linear (L) and quadratic (Q) Bayesian classifiers, with prior probabilities estimated empirically from the data and the Gaussian approximation of the class-conditional PDFs. The step sizes of the gradient descent in IDA, MIM and the proposed method are set to be 0.01. The experimental results are summarized in the following Tables 2-3, where the best performance for each m , i.e., a given size of subspace, is shown in bold and the best overall performance is shown in bold italic. The performances, represented by the classification accuracies and standard deviations, are shown in Tables 2-3.

For these two datasets, the proposed method significantly outperforms other methods. The reason is that when the size-dimension ratio is small, i.e., in the case of high-dimensional limited data, other methods cannot estimate these statistics accurately in the original high-dimensional space and their performances are generally weaker. In such cases, not only the estimation errors in the low-dimensional feature space can be reduced, but also the between-class separability can

be improved, the proposed method significantly outperforms other methods.

Comparing the linear classifiers with the quadratic classifiers, the linear classifier is superior to the quadratic classifier on Heart, while the quadratic classifier is superior to the linear classifier on Satellite. Therefore, whether a classifier can perform well on a dataset also depends on the specific characteristics of the dataset.

B. Radar Data

In this Section, we examine the performance of our method on the measured High Resolution Range Profile (HRRP) data. The measured HRRP data used in this paper are ISAR data based on C band radar, and the center frequency is 5520 MHz, the signal bandwidth is 400 MHz, the pulse repetition frequency is 400 Hz, the range resolution is 0.375 m, and the wavelength is 0.05 m. The radar targets are three airplanes, including a big yet plane Yark-42, a small yet plane Cessna Citation S/II, and a propeller plane An-26. The parameters of the airplane targets and measurement radar are shown in Table 4.

We present the projections of target trajectories onto the ground plane in Fig. 2, from which the aspect angle of the airplane can be estimated according to its relative position to the radar. As shown in Fig. 2, the measured data are segmented, and each segment contains 25600 HRRP samples, and the dimension of an HRRP is 256, i.e., 256 range cells.

TABLE II. AVERAGE CLASSIFICATION ACCURACIES (%) AND STANDARD DEVIATIONS (%) FOR HEART DATA

Method		Size of subspace (m)				
		1	2	3	4	5
FA	L	79.37±1.80	79.37±2.68	80.67±2.27	81.22±2.42	82.44±1.94
	Q	79.26±1.88	79.89±2.72	80.63±2.96	80.37±2.49	81.26±1.92
PCA	L	56.67±3.04	68.19±3.20	68.59±3.44	68.26±3.91	78.59±2.15
	Q	56.44±4.08	67.22±3.51	66.85±3.19	65.70±4.01	76.22±2.56
LDA	L	83.22±1.93	-	-	-	-
	Q	83.19±2.07	-	-	-	-
IDA	L	75.78±3.69	82.67±3.79	83.85±2.62	83.59±2.15	83.44±2.55
	Q	75.67±3.67	82.33±3.91	81.78±3.25	80.48±2.86	79.70±3.16
MIM	L	77.93±4.20	83.19±1.95	83.30±1.90	83.07±2.01	83.37±2.05
	Q	77.89±4.33	82.41±2.32	82.85±2.36	82.26±2.00	81.37±2.70
MMI-FA	L	84.52±1.46	84.22±1.75	84.11±2.34	84.33±2.44	83.85±2.36
	Q	84.78±1.55	84.70±1.72	84.22±2.07	83.93±2.44	83.41±2.71

TABLE III. AVERAGE CLASSIFICATION ACCURACIES (%) AND STANDARD DEVIATIONS (%) FOR SATELLITE DATA

Method		Size of subspace (m)				
		1	3	5	7	8
FA	L	57.75	80.95	80.30	80.95	81.35
	Q	57.80	84.25	83.35	84.25	84.05
PCA	L	50.90	80.50	80.40	81.00	81.25
	Q	51.10	83.70	84.10	83.80	83.75
LDA	L	52.35	82.30	82.85	-	-
	Q	56.50	84.15	84.45	-	-
IDA	L	59.30	82.00	82.05	81.90	81.80
	Q	67.30	84.70	83.80	83.75	83.85
MIM	L	71.55	81.55	81.20	81.45	81.50
	Q	72.65	83.70	84.10	83.85	83.95
MMI-FA	L	51.75	82.00	83.25	83.05	82.05
	Q	53.50	85.15	86.20	86.50	85.70

TABLE IV. PARAMETERS OF THE THREE AIRPLANES AND RADAR IN THE EXPERIMENT

Planes	Length (m)	Width (m)	Height (m)
Yark-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

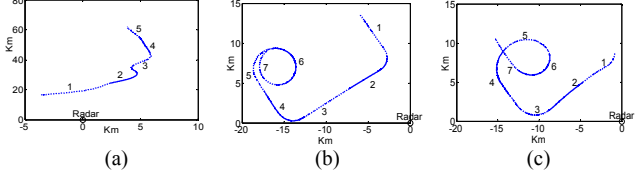


Fig. 2. Projections of target trajectories onto the ground plane. (a) Yark-42. (b) Cessna Citation S/II. (c) An-26.

Data from the second and a part of the fifth segments of Yark-42, the sixth and seventh segments of Cessna Citation S/II, and the fifth and the sixth segments of An-26 are taken as training samples, and other data segments are taken as test samples in our experiments. For each segment of the training data, we choose 50 HRRP samples at intervals. As the introduced in [14], 10-dimensional power spectrum features, 7-dimensional central moment features, 5-dimensional autoregressive coefficient features are extracted from each HRRP sample to contain the 22-dimensional original feature vector.

Same with the experiments on benchmark dataset, the results is averaged over 20 random partitions of training and test datasets. The performances, represented by the average values and standard deviations, are shown in Table 5. The model parameters' settings are the same as those in the experiments on benchmark datasets. We observe that the best performance in each column, i.e., for each m , is obtained by the proposed method in most cases.

IV. CONCLUSION

In this paper, a novel method for supervised linear dimensionality reduction is proposed. The proposed method describes the subspace statistical structure of the observed data via factor analysis model and simultaneously ensures the separability of the transformed data via MI. Moreover, the optimal transformation matrix and noise variance can be obtained via the joint optimization of MI function and log-

likelihood function. The proposed method is validated by comparing with FA, PCA, LDA, IDA, and MIM on Benchmark datasets and HRRP data. Based on the experimental results, the proposed method outperforms the traditional supervised dimensionality reduction methods. Therefore, we infer that the proposed method should be considered as an alternative to the prevalent feature transformation methods.

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TABLE V. AVERAGE CLASSIFICATION ACCURACIES (%) AND STANDARD DEVIATIONS (%) FOR HRRP DATA

Method		Size of subspace (m)				
		2	4	5	7	8
FA	L	69.44±4.30	78.37±1.16	77.81±2.03	80.52±2.15	81.69±2.25
	Q	71.18±4.53	78.12±2.62	78.60±3.03	81.45±1.73	82.54±2.04
PCA	L	72.35±0.55	84.02±1.28	84.62±1.51	88.00±1.46	89.13±0.90
	Q	69.48±1.73	88.47±1.24	88.34±1.11	90.66±1.02	90.91±0.91
LDA	L	91.12±0.65	-	-	-	-
	Q	92.51±1.64	-	-	-	-
IDA	L	72.54±2.81	91.42±0.77	91.31±1.05	90.10±1.58	90.34±1.32
	Q	82.49±3.35	92.52±1.11	92.09±1.19	91.09±1.35	91.08±1.22
MIM	L	82.17±3.07	87.57±1.24	88.89±0.98	89.52±1.01	90.44±0.90
	Q	88.29±2.28	91.13±1.02	92.39±0.76	92.71±0.75	92.18±1.12
MMI-FA	L	82.52±2.31	90.44±1.49	90.60±1.22	90.22±1.49	90.84±1.21
	Q	92.06±2.64	92.81±0.96	92.89±1.00	93.36±0.57	93.10±0.64